

Numerical Methods for Root Finding: Comprehensive Analysis

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Introduction

This report presents a comprehensive analysis of various numerical methods for finding roots of nonlinear equations. We compare different methods in terms of convergence speed, accuracy, stability, and computational efficiency. The methods analyzed include:

- Bisection Method
- Newton's Method
- Secant Method
- Fixed-point Method
- Aitken's Acceleration
- Real-time Aitken's Acceleration

Test Functions

We tested the methods on the following function:

Primary Test Function: $f(x) = 3\sin(x) + 4\cos(x)$

- Mathematical form:** $f(x) = 3\sin(x) + 4\cos(x)$
- Derivative:** $f'(x) = -4\sin(x) + 3\cos(x)$
- Interval:** $[1.5000000000000000, 3.0000000000000000]$
- Initial guess:** 2.0000000000000000
- True root:** 2.214297435588181
- Tolerance:** $\epsilon = 0.0000010000000000$

This function represents exponential growth problems and has applications in population dynamics, compound interest, and decay processes.

Theoretical Background

Convergence Rates

- Linear Convergence:** If $|e_{n+1}| \leq c \cdot |e_n|$ for some constant $c < 1$, where e_n is the error at iteration n . The bisection method exhibits linear convergence.
- Quadratic Convergence:** If $|e_{n+1}| \leq c \cdot |e_n|^2$ for some constant c . Newton's method shows quadratic convergence when the function is well-behaved.
- Superlinear Convergence:** If $|e_{n+1}| \leq c \cdot |e_n|^p$ for $p > 1$. The secant method typically has $p \approx 1.62$.

Stopping Criteria

- Absolute Error:** $|x_{n+1} - x_n| < \epsilon$
- Relative Error:** $|x_{n+1} - x_n| / |x_{n+1}| < \epsilon$
- Function Value:** $|f(x_n)| < \epsilon$

Aitken's Acceleration

Aitken's acceleration transforms a linearly convergent sequence into a more rapidly convergent one. For a sequence $\{p_n\}$, the Aitken's accelerated sequence $\{\hat{p}_n\}$ is defined as:

$$\hat{p}_n = p_n - (p_{n+1} - p_n)^2 / (p_{n+2} - 2p_{n+1} + p_n)$$

Method Descriptions with Examples

1. Bisection Method

Mathematical Formula

Given a continuous function $f(x)$ and an interval $[a, b]$ where $f(a) \cdot f(b) < 0$:

$$x = a + b/2$$

The root lies in $[a, c]$ if $f(a) \cdot f(x) < 0$, otherwise in $[x, b]$.

Algorithm

- Input:** Function f , interval $[a, b]$, tolerance ϵ
- Verify:** Check that $f(a) \cdot f(b) < 0$
- Iterate:**
- Compute midpoint: $x = a + b/2$
- If $|f(x)| < \epsilon$, return x
- If $f(a) \cdot f(x) < 0$, set $b = x$
- Otherwise, set $a = x$
- Repeat** until convergence

Convergence Rate

Linear convergence with rate ≈ 0.5 . Error reduces by half each iteration: $|e_{n+1}| \leq 1/2 |e_n|$

Implementation

```
def bisection_method(f, a, b, epsilon=1e-6, condition=stopping_condition_2):
    if f(a) * f(b) > 0:
        raise ValueError(f"f({a}) and f({b}) must have opposite signs")

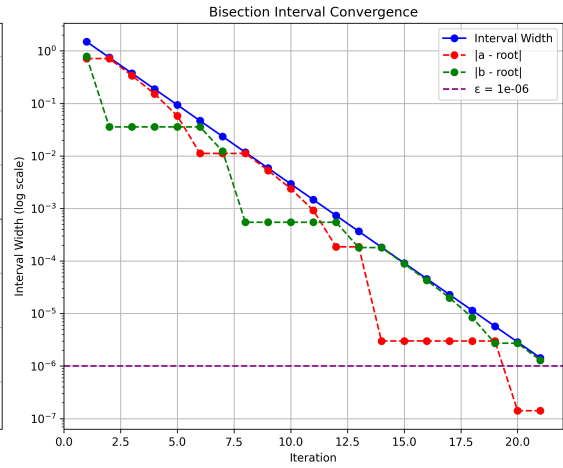
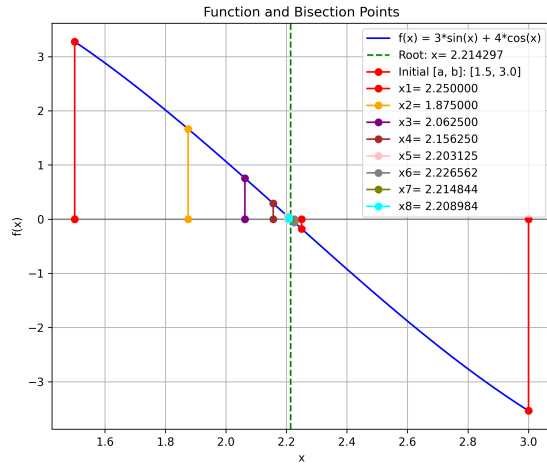
    sequence = []
    for k in range(100):
        x = (a + b) / 2
        sequence.append(x)

        if k > 0 and condition(sequence[k], sequence[k - 1], epsilon):
            return sequence, "Second stopping condition (convergence)"

    if f(a) * f(x) < 0:
        b = x
    else:
        a = x
```

return sequence, "First stopping condition (max iterations)"

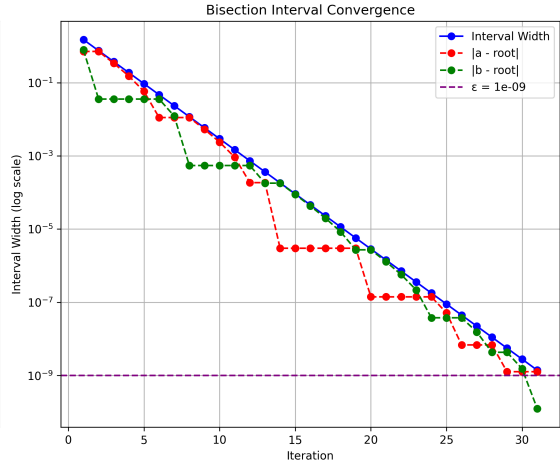
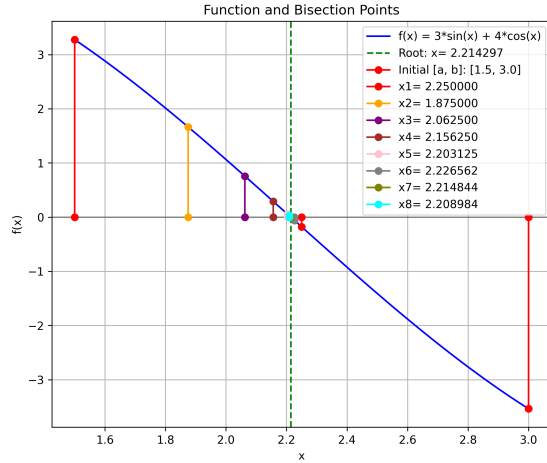
Examples with Absolute Error Criterion $|x_{n+1} - x_n| < \epsilon$



Bisection Method Absolute Criterion ($\epsilon = 0.0000010000000000$)

Iteration	x_n	$f(x_n)$	Error	Interval $[a, b]$	Interval Length	Abs. Difference	Aitken
1	2.2500000000000000	-0.178474900227193	0.035702564411819	[1.5000000000000000, 3.0000000000000000]	1.5000000000000000	N/A	N/A
2	1.8750000000000000	1.664123320070785	0.339297435588181	[1.5000000000000000, 2.2500000000000000]	0.7500000000000000	0.3750000000000000	N/A
3	2.0625000000000000	0.756075712201294	0.151797435588181	[1.8750000000000000, 2.2500000000000000]	0.3750000000000000	0.1875000000000000	2.0
4	2.1562500000000000	0.290074212805930	0.058047435588181	[2.0625000000000000, 2.2500000000000000]	0.1875000000000000	0.0937500000000000	2.25
5	2.2031250000000000	0.055861015797768	0.011172435588181	[2.1562500000000000, 2.2500000000000000]	0.0937500000000000	0.0468750000000000	2.25
6	2.2265625000000000	-0.061323784524178	0.012265064411819	[2.2031250000000000, 2.2500000000000000]	0.0468750000000000	0.0234375000000000	2.25
7	2.2148437500000000	-0.002731571923218	0.000546314411819	[2.2031250000000000, 2.2265625000000000]	0.0234375000000000	0.0117187500000000	2.21875
8	2.2089843750000000	0.026565177957472	0.005313060588181	[2.2031250000000000, 2.2148437500000000]	0.0117187500000000	0.0058593750000000	2.203125
9	2.2119140625000000	0.011916854158681	0.002383373088181	[2.2089843750000000, 2.2148437500000000]	0.0058593750000000	0.0029296875000000	2.2109375
10	2.2133789062500000	0.004592646045105	0.000918529338181	[2.2119140625000000, 2.2148437500000000]	0.0029296875000000	0.0014648437500000	2.21484375
11	2.2141113281250000	0.000930537310534	0.000186107463181	[2.2133789062500000, 2.2148437500000000]	0.0014648437500000	0.0007324218750000	2.21484375
12	2.2144775390625000	-0.000900517366726	0.000180103474319	[2.2141113281250000, 2.2148437500000000]	0.0007324218750000	0.0003662109375000	2.21484375
13	2.2142944335937500	0.000015009972155	0.000003001994431	[2.2141113281250000, 2.2144775390625000]	0.0003662109375000	0.0001831054687500	2.21435546875
14	2.2143859863281250	-0.000442753699141	0.000088550739944	[2.2142944335937500, 2.2144775390625000]	0.0001831054687500	0.0000915527343750	2.21435546875
15	2.2143402099609380	-0.000213871863717	0.000042774372757	[2.2142944335937500, 2.2143859863281250]	0.0000915527343750	0.0000457763671880	2.21435546875
16	2.2143173217773440	-0.000099430945807	0.000019886189163	[2.2142944335937500, 2.2143402099609380]	0.0000457763671880	0.0000228881835940	2.21429443359375
17	2.2143058776855470	-0.000042210486829	0.000008442097366	[2.2142944335937500, 2.2143173217773440]	0.0000228881835940	0.0000114440917970	2.21429443359375
18	2.2143001556396480	-0.000013600257337	0.000002720051468	[2.2142944335937500, 2.2143058776855470]	0.0000114440917970	0.0000057220458980	2.21429443359375
19	2.2142972946166990	0.000000704857409	0.000000140971482	[2.2142944335937500, 2.2143001556396480]	0.0000057220458980	0.0000028610229490	2.21429443359375
20	2.2142987251281740	-0.000006447699964	0.000001289539993	[2.2142972946166990, 2.2143001556396480]	0.0000028610229490	0.0000014305114750	2.2142982482910156
21	2.2142980098724370	-0.000002871421278	0.000000574284256	[2.2142972946166990, 2.2142987251281740]	0.0000014305114750	0.0000007152557370	2.2142982482910156

Result: Converged to 2.214298009872437 in 21 iterations **Final Error:** 0.000000574284256 **Convergence Rate:** 1.322232053655269

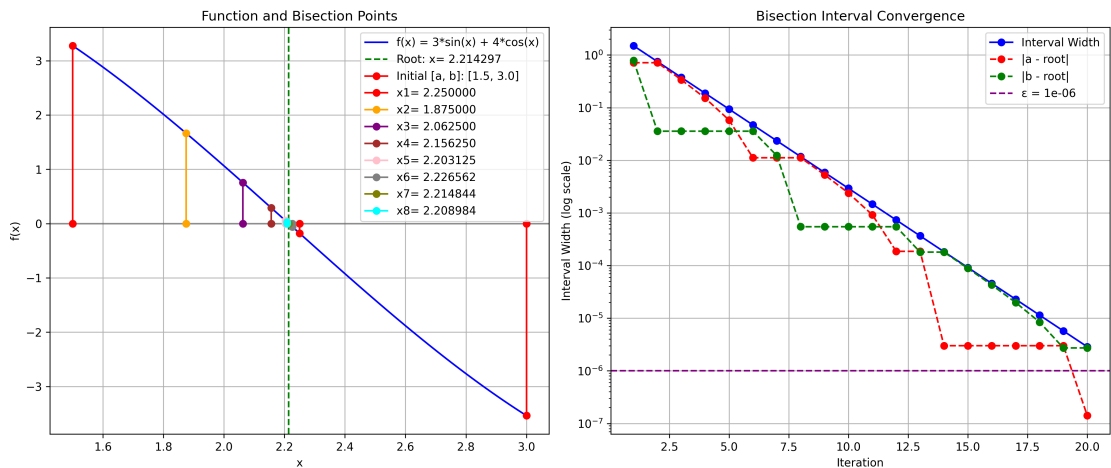


Bisection Method Absolute Criterion ($\epsilon = 0.000000001000000$)

Iteration	x_n	$f(x_n)$	Error	Interval $[a, b]$	Interval Length	Abs. Difference	Aitken
1	2.2500000000000000	-0.178474900227193	0.035702564411819	[1.5000000000000000, 3.0000000000000000]	1.5000000000000000	N/A	N/A
2	1.8750000000000000	1.664123320070785	0.339297435588181	[1.5000000000000000, 2.2500000000000000]	0.7500000000000000	0.3750000000000000	N/A
3	2.0625000000000000	0.756075712201294	0.151797435588181	[1.8750000000000000, 2.2500000000000000]	0.3750000000000000	0.1875000000000000	2.0
4	2.1562500000000000	0.290074212805930	0.058047435588181	[2.0625000000000000, 2.2500000000000000]	0.1875000000000000	0.0937500000000000	2.25
5	2.2031250000000000	0.055861015797768	0.011172435588181	[2.1562500000000000, 2.2500000000000000]	0.0937500000000000	0.0468750000000000	2.25
6	2.2265625000000000	-0.061323784524178	0.012265064411819	[2.2031250000000000, 2.2500000000000000]	0.0468750000000000	0.0234375000000000	2.25
7	2.2148437500000000	-0.002731571923218	0.000546314411819	[2.2031250000000000, 2.2265625000000000]	0.0234375000000000	0.0117187500000000	2.21875
8	2.2089843750000000	0.026565177957472	0.005313060588181	[2.2031250000000000, 2.2148437500000000]	0.0117187500000000	0.0058593750000000	2.203125
9	2.2119140625000000	0.011916854158681	0.002383373088181	[2.2089843750000000, 2.2148437500000000]	0.0058593750000000	0.0029296875000000	2.2109375
10	2.2133789062500000	0.004592646045105	0.000918529338181	[2.2119140625000000, 2.2148437500000000]	0.0029296875000000	0.0014648437500000	2.21484375
11	2.2141113281250000	0.000930537310534	0.000186107463181	[2.2133789062500000, 2.2148437500000000]	0.0014648437500000	0.0007324218750000	2.21484375
12	2.2144775390625000	-0.000900517366726	0.000180103474319	[2.2141113281250000, 2.2144775390625000]	0.0007324218750000	0.0003662109375000	2.21484375
13	2.2142944335937500	0.000015009972155	0.00003001994431	[2.2141113281250000, 2.2144775390625000]	0.0003662109375000	0.0001831054687500	2.21435546875
14	2.2143859863281250	-0.000442753699141	0.000088550739944	[2.2142944335937500, 2.2144775390625000]	0.0001831054687500	0.0000915527343750	2.21435546875
15	2.2143402099609380	-0.000213871863717	0.000042774372757	[2.2142944335937500, 2.2143859863281250]	0.0000915527343750	0.0000457763671880	2.21435546875
16	2.2143173217773440	-0.000099430945807	0.000019886189163	[2.2142944335937500, 2.2143402099609380]	0.0000457763671880	0.0000228881835940	2.21429443359375
17	2.2143058776855470	-0.000042210486829	0.000008442097366	[2.2142944335937500, 2.2143173217773440]	0.0000228881835940	0.0000114440917970	2.21429443359375
18	2.2143001556396480	-0.000013600257337	0.000002720051468	[2.2142944335937500, 2.2143058776855470]	0.0000114440917970	0.0000057220458980	2.21429443359375
19	2.2142972946166990	0.000000704857409	0.000000140971482	[2.2142944335937500, 2.2143001556396480]	0.0000057220458980	0.0000028610229490	2.21429443359375
20	2.2142987251281740	-0.000006447699964	0.000001289539993	[2.2142972946166990, 2.2143001556396480]	0.0000028610229490	0.0000014305114750	2.2142982482910156
21	2.2142980098724370	-0.000002871421278	0.000000574284256	[2.2142972946166990, 2.2142987251281740]	0.0000014305114750	0.0000007152557370	2.2142982482910156
22	2.2142976522445680	-0.000001083281934	0.000000216656387	[2.2142972946166990, 2.2142980098724370]	0.0000007152557370	0.0000003576278690	2.2142972946166990
23	2.2142974734306340	-0.000000189212263	0.000000037842453	[2.2142972946166990, 2.2142976522445680]	0.0000003576278690	0.0000001788139340	2.2142972946166990
24	2.2142973840236660	0.0000000257822573	0.000000051564514	[2.2142972946166990, 2.2142974734306340]	0.0000001788139340	0.0000000894069670	2.2142972946166990
25	2.2142974287271500	0.000000034305155	0.000000006861031	[2.2142973840236660, 2.2142974734306340]	0.0000000894069670	0.0000000447034840	2.2142974138259888
26	2.2142974510788920	-0.000000077453553	0.000000015490711	[2.2142974287271500, 2.2142974734306340]	0.0000000447034840	0.0000000223517420	2.2142974734306335
27	2.2142974399030210	-0.000000021574199	0.000000004314840	[2.2142974287271500, 2.2142974510788920]	0.0000000223517420	0.0000000117587100	2.2142974436283110
28	2.2142974343150850	0.000000006365478	0.000000001273095	[2.2142974399030210, 2.2142974510788920]	0.0000000117587100	0.0000000055879350	2.2142974287271500
29	2.2142974371090530	-0.000000007604361	0.000000001520872	[2.2142974343150850, 2.2142974399030210]	0.0000000055879350	0.0000000027939680	2.2142974361777306
30	2.2142974357120690	-0.000000000619441	0.000000000123888	[2.2142974371090530, 2.2142974399030210]	0.0000000027939680	0.0000000013969840	2.2142974361777306
31	2.2142974350135770	0.0000000002873018	0.000000000574603	[2.2142974343150850, 2.2142974357120690]	0.0000000013969840	0.0000000006984920	2.2142974343150854

Result: Converged to 2.214297435013577 in 31 iterations **Final Error:** 0.00000000574603 **Convergence Rate:** 1.320841459834379

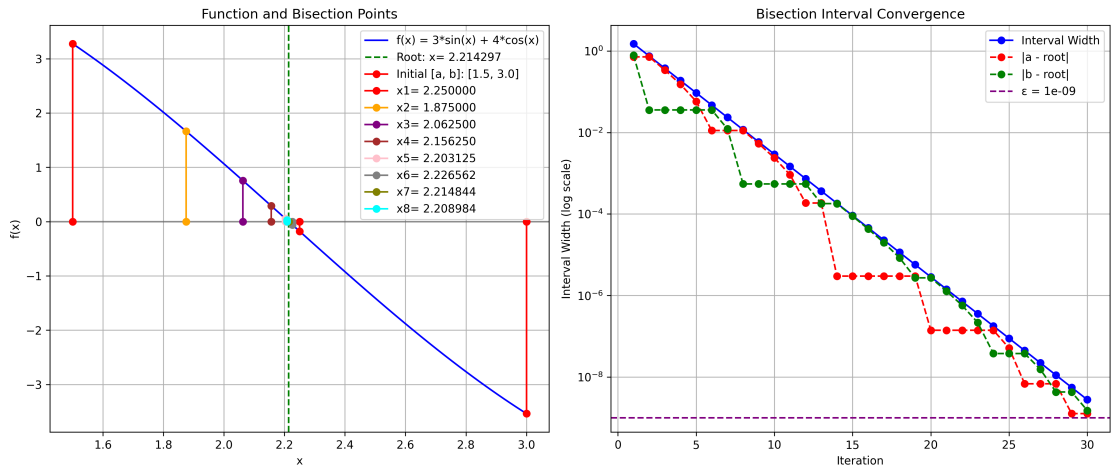
Examples with Relative Error Criterion $|x_{n+1} - x_n|/|x_{n+1}| < \epsilon$



Bisection Method with Relative Criterion ($\epsilon = 0.000001000000000$)

Iteration	x_n	$f(x_n)$	Error	Interval [a, b]	Interval Length	Rel. Difference	Aitken
1	2.250000000000000	-0.178474900227193	0.035702564411819	[1.500000000000000, 3.000000000000000]	1.500000000000000	N/A	N/A
2	1.875000000000000	1.664123320070785	0.339297435588181	[1.500000000000000, 2.250000000000000]	0.750000000000000	0.200000000000000	N/A
3	2.062500000000000	0.756075712201294	0.151797435588181	[1.875000000000000, 2.250000000000000]	0.375000000000000	0.090909090909091	2.0
4	2.156250000000000	0.290074212805930	0.058047435588181	[2.062500000000000, 2.250000000000000]	0.187500000000000	0.043478260869565	2.25
5	2.203125000000000	0.055861015797768	0.011172435588181	[2.156250000000000, 2.250000000000000]	0.093750000000000	0.021276595744681	2.25
6	2.226562500000000	-0.061323784524178	0.012265064411819	[2.203125000000000, 2.250000000000000]	0.046875000000000	0.010526315789474	2.25
7	2.214843750000000	-0.002731571923218	0.000546314411819	[2.203125000000000, 2.226562500000000]	0.023437500000000	0.005291005291005	2.21875
8	2.208984375000000	0.026565177957472	0.005313060588181	[2.203125000000000, 2.214843750000000]	0.011718750000000	0.002652519893899	2.203125
9	2.211914062500000	0.011916854158681	0.002383373088181	[2.208984375000000, 2.214843750000000]	0.005859375000000	0.001324503311258	2.2109375
10	2.213378906250000	0.004592646045105	0.000918529338181	[2.211914062500000, 2.214843750000000]	0.002929687500000	0.000661813368630	2.21484375
11	2.214111328125000	0.000930537310534	0.000186107463181	[2.213378906250000, 2.214843750000000]	0.001464843750000	0.000330797221303	2.21484375
12	2.214477539062500	-0.000900517366726	0.000180103474319	[2.214111328125000, 2.214843750000000]	0.000732421875000	0.000165371258475	2.21484375
13	2.214294433593750	0.000015009972155	0.00003001994431	[2.214111328125000, 2.214477539062500]	0.000366210937500	0.000082692466716	2.21435546875
14	2.214385986328125	-0.000442753699141	0.000088550739944	[2.214294433593750, 2.214477539062500]	0.000183105468750	0.000041344523918	2.21435546875
15	2.214340209960938	-0.000213871863717	0.000042774372757	[2.214294433593750, 2.214385986328125]	0.000091552734375	0.000020672689310	2.21435546875
16	2.214317321777344	-0.000099430945807	0.000019886189163	[2.214294433593750, 2.214340209960938]	0.000045776367188	0.000010336451496	2.21429443359375
17	2.214305877685547	-0.000042210486829	0.000008442097366	[2.214294433593750, 2.214317321777344]	0.000022888183594	0.000005168252459	2.21429443359375
18	2.214300155639648	-0.000013600257337	0.000002720051468	[2.214294433593750, 2.214305877685547]	0.000011444091797	0.000002584132907	2.21429443359375
19	2.214297294616699	0.000000704857409	0.000000140971482	[2.214294433593750, 2.214300155639648]	0.000005722045898	0.000001292068123	2.21429443359375
20	2.214298725128174	-0.000006447699964	0.000001289539993	[2.214297294616699, 2.214300155639648]	0.000002861022949	0.000000646033644	2.2142982482910156

Result: Converged to 2.214298725128174 in 20 iterations Final Error: 0.000001289539993 Convergence Rate: 1.322232053655269

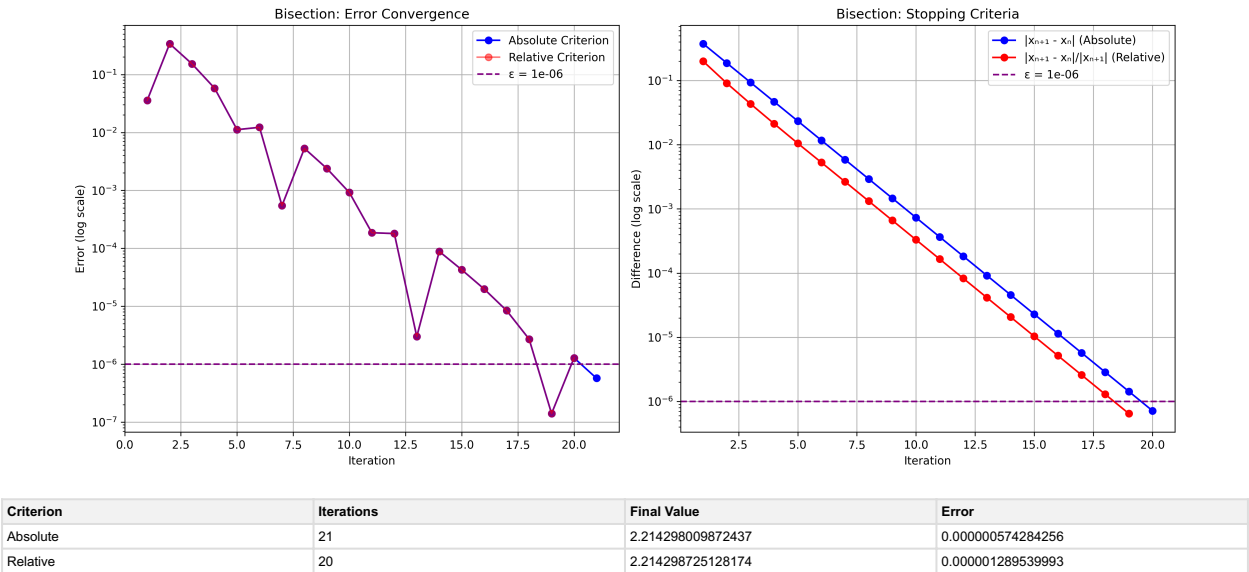


Bisection Method with Relative Criterion ($\epsilon = 0.00000001000000$)

Iteration	x_n	$f(x_n)$	Error	Interval [a, b]	Interval Length	Rel. Difference	Aitken
1	2.250000000000000	-0.178474900227193	0.035702564411819	[1.500000000000000, 3.000000000000000]	1.500000000000000	N/A	N/A
2	1.875000000000000	1.664123320070785	0.339297435588181	[1.500000000000000, 2.250000000000000]	0.750000000000000	0.200000000000000	N/A
3	2.062500000000000	0.756075712201294	0.151797435588181	[1.875000000000000, 2.250000000000000]	0.375000000000000	0.090909090909091	2.0
4	2.156250000000000	0.290074212805930	0.058047435588181	[2.062500000000000, 2.250000000000000]	0.187500000000000	0.043478260869565	2.25
5	2.203125000000000	0.055861015797768	0.011172435588181	[2.156250000000000, 2.250000000000000]	0.093750000000000	0.021276595744681	2.25
6	2.226562500000000	-0.061323784524178	0.012265064411819	[2.203125000000000, 2.250000000000000]	0.046875000000000	0.010526315789474	2.25
7	2.214843750000000	-0.002731571923218	0.000546314411819	[2.203125000000000, 2.226562500000000]	0.023437500000000	0.005291005291005	2.21875
8	2.208984375000000	0.026565177957472	0.005313060588181	[2.203125000000000, 2.214843750000000]	0.011718750000000	0.002652519893899	2.203125
9	2.211914062500000	0.011916854158681	0.002383373088181	[2.208984375000000, 2.214843750000000]	0.005859375000000	0.001324503311258	2.2109375
10	2.213378906250000	0.004592646045105	0.000918529338181	[2.211914062500000, 2.214843750000000]	0.002929687500000	0.000661813368630	2.21484375
11	2.214111328125000	0.000930537310534	0.000186107463181	[2.213378906250000, 2.214843750000000]	0.001464843750000	0.000330797221303	2.21484375
12	2.214477539062500	-0.000900517366726	0.000180103474319	[2.214111328125000, 2.214843750000000]	0.000732421875000	0.000165371258475	2.21484375
13	2.214294433593750	0.000015009972155	0.000003001994431	[2.214111328125000, 2.214477539062500]	0.000366210937500	0.000082692466716	2.21435546875
14	2.214385986328125	-0.000442753699141	0.000088550739944	[2.214294433593750, 2.214477539062500]	0.000183105468750	0.000041344523918	2.21435546875
15	2.214340209960938	-0.000213871863717	0.000042774372757	[2.214294433593750, 2.214385986328125]	0.000091552734375	0.000020672689310	2.21435546875
16	2.214317321777344	-0.000099430945807	0.000019886189163	[2.214294433593750, 2.214340209960938]	0.000045776367188	0.000010336451496	2.21429443359375
17	2.214305877685547	-0.000042210486829	0.000008442097366	[2.214294433593750, 2.214317321777344]	0.000022888183594	0.000005168252459	2.21429443359375
18	2.214300155639648	-0.000013600257337	0.000002720051468	[2.214294433593750, 2.214305877685547]	0.000011444091797	0.000002584132907	2.21429443359375
19	2.214297294616699	0.000000704857409	0.000000140971482	[2.214294433593750, 2.214300155639648]	0.000005722045898	0.000001292068123	2.21429443359375
20	2.214298725128174	-0.000006447699964	0.000001289539993	[2.214297294616699, 2.214300155639648]	0.000002861022949	0.000000646033644	2.2142982482910156
21	2.214298009872437	-0.000002871421278	0.000000574284256	[2.214297294616699, 2.214298725128174]	0.000001430511475	0.000000323016926	2.2142982482910156
22	2.214297652244568	-0.000001083281934	0.000000216656387	[2.214297294616699, 2.214298009872437]	0.000000715255737	0.000000161508489	2.214297294616699
23	2.214297473430634	-0.000000189212263	0.000000037842453	[2.214297294616699, 2.214297652244568]	0.000000357627869	0.000000080754251	2.214297294616699
24	2.214297384023666	0.0000000257822573	0.000000051564514	[2.214297294616699, 2.214297473430634]	0.000000178813934	0.000000040377127	2.214297294616699
25	2.214297428727150	0.000000034305155	0.000000006861031	[2.214297384023666, 2.214297473430634]	0.000000089406967	0.000000020188563	2.2142974138259888
26	2.214297451078892	-0.000000077453553	0.000000015490711	[2.214297428727150, 2.214297473430634]	0.000000044703484	0.000000010094281	2.2142974734306335
27	2.214297439903021	-0.000000021574199	0.000000004314840	[2.214297428727150, 2.214297451078892]	0.000000022351742	0.000000005047141	2.214297443628311
28	2.214297434315085	0.000000006365478	0.000000001273095	[2.214297428727150, 2.214297439903021]	0.00000001175871	0.000000002523570	2.21429742872715
29	2.214297437109053	-0.000000007604361	0.000000001520872	[2.214297434315085, 2.214297439903021]	0.000000005587935	0.000000001261785	2.2142974361777306
30	2.214297435712069	-0.00000000619441	0.00000000123888	[2.214297434315085, 2.214297437109053]	0.000000002793968	0.000000000630893	2.2142974361777306

Result: Converged to 2.214297435712069 in 30 iterations **Final Error:** 0.00000000123888 **Convergence Rate:** 1.320841459834379

Comparison of Stopping Criteria for Bisection Method



2. Newton's Method

Mathematical Formula

Starting with an initial guess x_0 , the Newton-Raphson iteration is:

$$x_{n+1} = x_n - f(x_n)/f'(x_n)$$

This formula comes from the linear approximation of $f(x)$ at x_n : $f(x) \approx f(x_n) + f'(x_n)(x - x_n)$

Setting $f(x) = 0$ and solving for x gives the Newton iteration.

Algorithm

1. **Input:** Function f , derivative f' , initial guess x_0 , tolerance ϵ
2. **Initialize:** Set $n = 0$
3. **Iterate:**
4. Compute $x_{n+1} = x_n - f(x_n)/f'(x_n)$
5. If $|error| < \epsilon$, return x_{n+1}
6. Set $n = n + 1$
7. **Repeat** until convergence or maximum iterations

Convergence Rate

Quadratic convergence when $f'(r) \neq 0$ at the root r : $|e_{n+1}| \leq C|e_n|^2$

where $C = |f''(r)/2f'(r)|$

Implementation

```
def newton_method(f, df, x0, epsilon=1e-6, condition=stopping_condition_2):
    sequence = [x0]
    x = x0

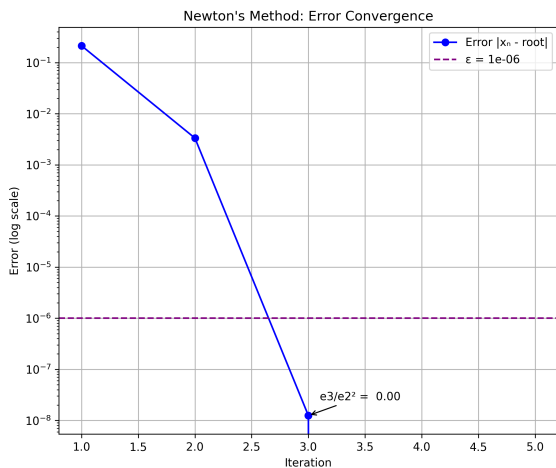
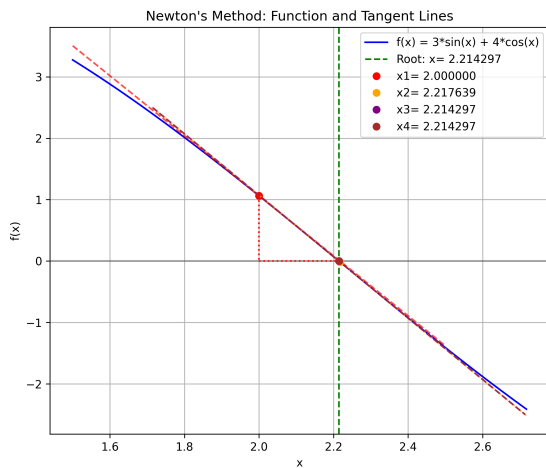
    for k in range(100):
        try:
            x_new = x - f(x) / df(x)
            sequence.append(x_new)

            if condition(sequence[k], sequence[k - 1], epsilon):
                return sequence, "Second stopping condition (convergence)"

            x = x_new
        except ZeroDivisionError:
            return sequence, "Error: Zero derivative"

    return sequence, "First stopping condition (max iterations)"
```

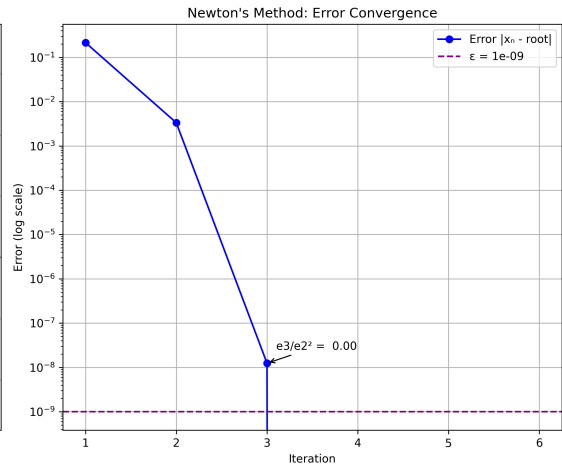
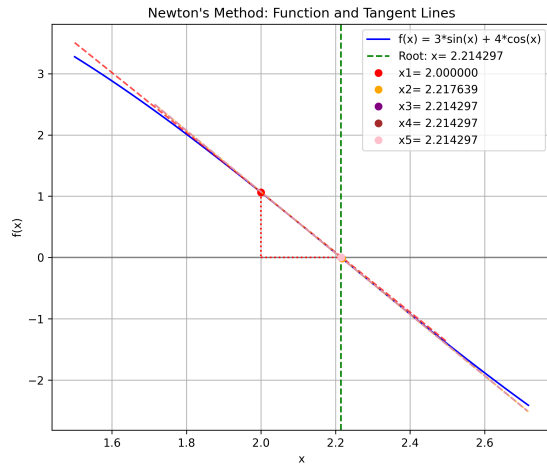
Examples with Absolute Error Criterion $|x_{n+1} - x_n| < \epsilon$



Newton's Method ($\epsilon = 0.000001000000000$)

Iteration	x_n	$f(x_n)$	$f'(x_n)$	Error	Abs. Error	Aitken
1	2.000000000000000	1.063304934288475	7.389056098930650	0.214297435588181	N/A	N/A
2	2.217639257797441	-0.016709079945716	9.185620367861599	0.003341822209260	0.217639257797441	N/A
3	2.214297423147885	0.00000062201478	9.154974778205764	0.00000012440295	0.003341834649556	2.2143479607694583
4	2.214297435588181	0.000000000000001	9.154974892096355	0.000000000000000	0.000000012440295	2.214297435588134
5	2.214297435588181	0.000000000000001	9.154974892096355	0.000000000000000	0.000000000000000	2.2142974355881804

Result: Converged to 2.214297435588181 in 5 iterations **Final Error:** 0.000000000000000 **Convergence Rate:** 3.004455781252174

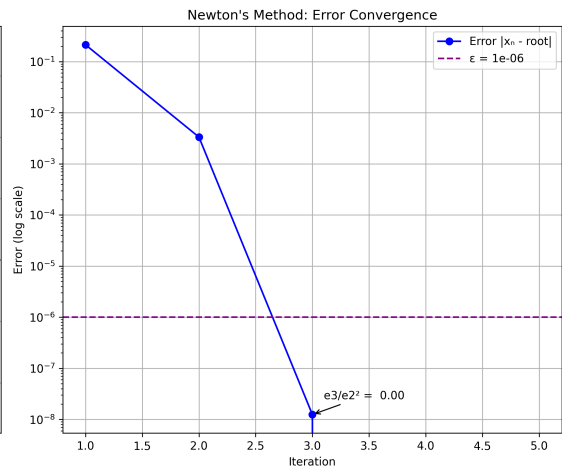
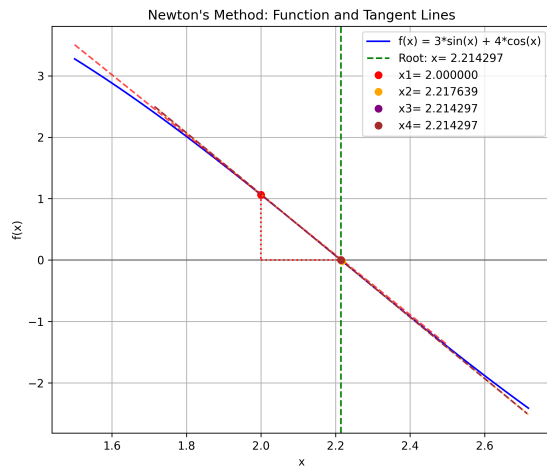


Newton's Method ($\epsilon = 0.00000001000000$)

Iteration	x_n	$f(x_n)$	$f'(x_n)$	Error	Abs. Error	Aitken
1	2.000000000000000	1.063304934288475	7.389056098930650	0.214297435588181	N/A	N/A
2	2.217639257797441	-0.016709079945716	9.185620367861599	0.003341822209260	0.217639257797441	N/A
3	2.214297423147885	0.00000062201478	9.154974778205764	0.000000012440295	0.003341834649556	2.2143479607694583
4	2.214297435588181	0.000000000000001	9.154974892096355	0.000000000000000	0.000000012440295	2.214297435588134
5	2.214297435588181	0.000000000000001	9.154974892096355	0.000000000000000	0.000000000000000	2.2142974355881804
6	2.214297435588181	0.000000000000001	9.154974892096355	0.000000000000000	0.000000000000000	2.214297435588181

Result: Converged to 2.214297435588181 in 6 iterations **Final Error:** 0.000000000000000 **Convergence Rate:** 3.004455781252174

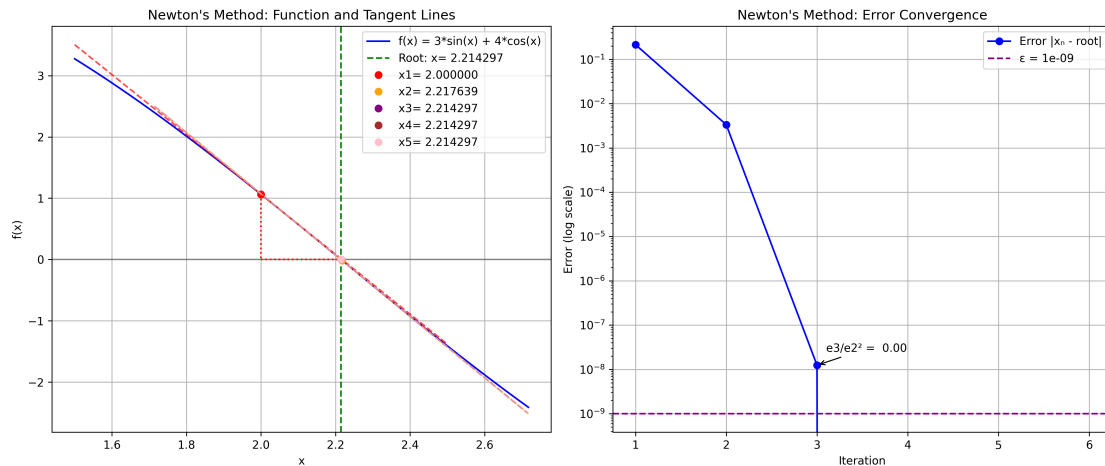
Examples with Relative Error Criterion $|x_{n+1} - x_n| / |x_{n+1}| < \epsilon$



Newton's Method with Relative Criterion ($\epsilon = 0.000001000000000$)

Iteration	x_n	$f(x_n)$	$f'(x_n)$	Error	Rel. Difference	Aitken
1	2.000000000000000	1.063304934288475	-4.885630216944154	0.214297435588181	N/A	N/A
2	2.217639257797441	-0.016709079945716	-4.999972080586787	0.003341822209260	0.098140063597900	N/A
3	2.214297423147885	0.00000062201478	-5.000000000000000	0.000000012440295	0.001509207667688	2.2143479607694583
4	2.214297435588181	0.000000000000001	-5.000000000000000	0.000000000000000	0.000000005618168	2.214297435588134
5	2.214297435588181	0.000000000000001	-5.000000000000000	0.000000000000000	0.000000000000000	2.2142974355881804

Result: Converged to 2.214297435588181 in 5 iterations **Final Error:** 0.000000000000000 **Convergence Rate:** 3.004455781252174

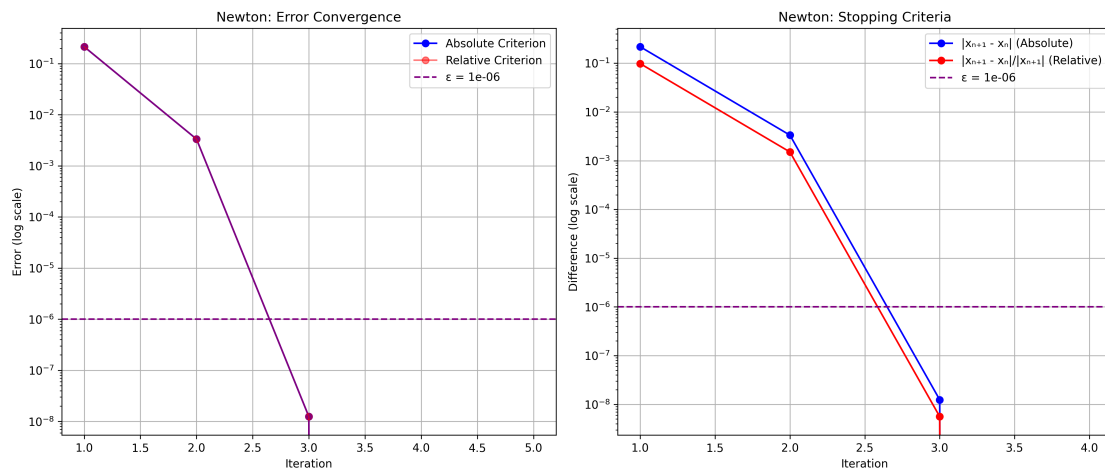


Newton's Method with Relative Criterion ($\epsilon = 0.00000001000000$)

Iteration	x_n	$f(x_n)$	$f'(x_n)$	Error	Rel. Difference	Aitken
1	2.000000000000000	1.063304934288475	-4.885630216944154	0.214297435588181	N/A	N/A
2	2.217639257797441	-0.016709079945716	-4.999972080586787	0.003341822209260	0.098140063597900	N/A
3	2.214297423147885	0.000000062201478	-5.000000000000000	0.00000012440295	0.001509207667688	2.2143479607694583
4	2.214297435588181	0.000000000000001	-5.000000000000000	0.000000000000000	0.000000005618168	2.214297435588134
5	2.214297435588181	0.000000000000001	-5.000000000000000	0.000000000000000	0.000000000000000	2.2142974355881804
6	2.214297435588181	0.000000000000001	-5.000000000000000	0.000000000000000	0.000000000000000	2.214297435588181

Result: Converged to 2.214297435588181 in 6 iterations **Final Error:** 0.000000000000000 **Convergence Rate:** 3.004455781252174

Comparison of Stopping Criteria for Newton's Method



Criterion	Iterations	Final Value	Error
Absolute	5	2.214297435588181	0.000000000000000
Relative	5	2.214297435588181	0.000000000000000

3. Secant Method

Mathematical Formula

The secant method approximates the derivative using two previous points:

$$x_{n+1} = x_n - f(x_n) \cdot \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

This can be rewritten as: $x_{n+1} = \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}$

Algorithm

- Input:** Function f , two initial points x_0, x_1 , tolerance ϵ
- Initialize:** Set $n = 1$
- Iterate:**
- Compute $x_{n+1} = x_n - f(x_n) \cdot \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$
- If $|error| < \epsilon$, return x_{n+1}
- Set $n = n + 1$
- Repeat** until convergence or maximum iterations

Convergence Rate

Superlinear convergence with rate $\phi = \frac{1 + \sqrt{5}}{2} \approx 1.618$ (golden ratio): $|e_{n+1}| \leq C|e_n|^\phi$

Implementation

```
def secant_method(f, x0, x1, epsilon=1e-6, condition=stopping_condition_2):
    sequence = []
    for k in range(100):
        try:
            f0, f1 = f(x0), f(x1)
            if abs(f1 - f0) < 1e-19:
                return sequence, "Error: Division by zero"
```

```

x_new = x1 - f1 * (x1 - x0) / (f1 - f0)
sequence.append(x_new)

if k > 0 and condition(sequence[k], sequence[k - 1], epsilon):
    return sequence, "Second stopping condition (convergence)"

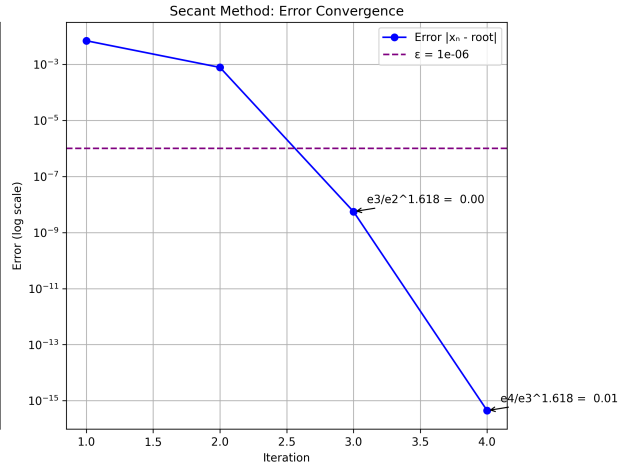
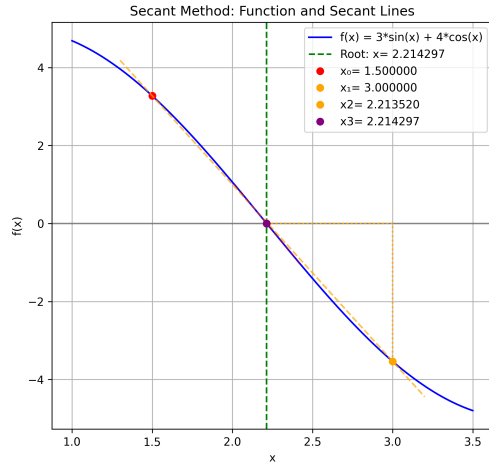
x0, x1 = x1, x_new

except (ZeroDivisionError, OverflowError):
    return sequence, "Error: Non-convergence"

return sequence, "First stopping condition (max iterations)"

```

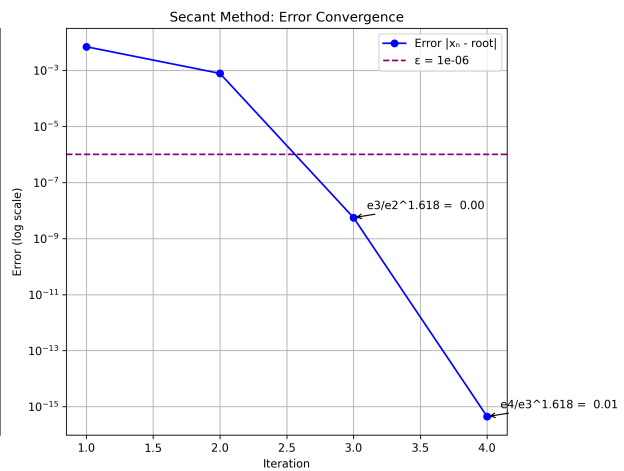
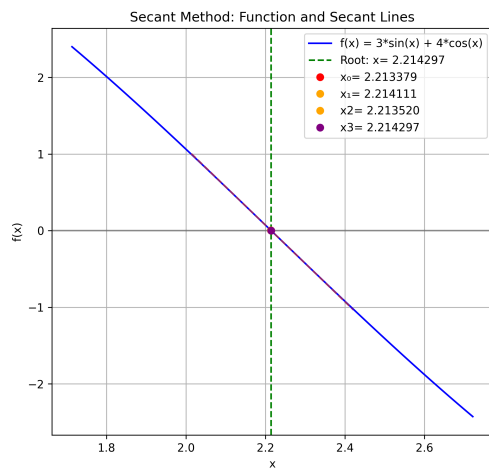
Examples with Absolute Error Criterion $|x_{n+1} - x_n| < \epsilon$



$f(x) = 3\sin(x) + 4\cos(x)$ ($\epsilon = 1e-06$, absolute criterion)

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
1	2.221244731448364	-0.034736199876025	0.006947295860183	N/A	N/A
2	2.213520010199476	0.003887126551967	0.000777425388705	0.007724721248888	N/A
3	2.214297441142124	-0.000000027769715	0.000000005553943	0.000777430942648	2.2142263533919904
4	2.214297435588180	0.0000000000000003	0.000000000000000	0.000000005553944	2.21429743558822

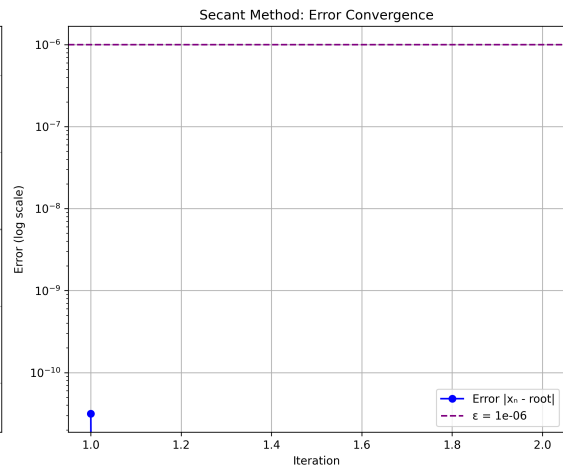
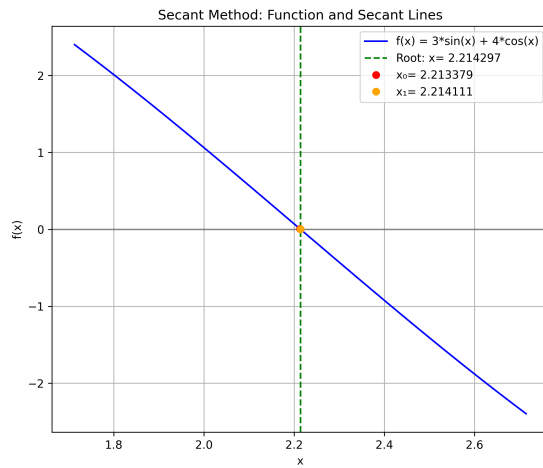
Result: Converged to 2.214297435588180 in 4 iterations **Final Error:** 0.000000000000000 **Convergence Rate:** 5.410312797770888



Secant method from bisection $[a, b]$ ($\epsilon = 1e-06$, absolute)

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
1	2.221244731448364	-0.034736199876025	0.006947295860183	N/A	N/A
2	2.213520010199476	0.003887126551967	0.000777425388705	0.007724721248888	N/A
3	2.214297441142124	-0.000000027769715	0.000000005553943	0.000777430942648	2.2142263533919904
4	2.214297435588180	0.0000000000000003	0.000000000000000	0.000000005553944	2.21429743558822

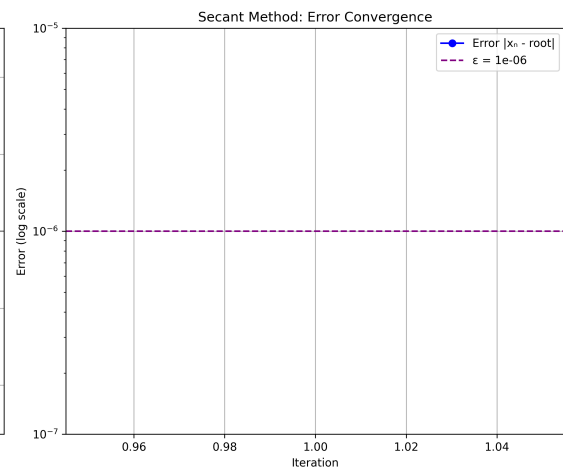
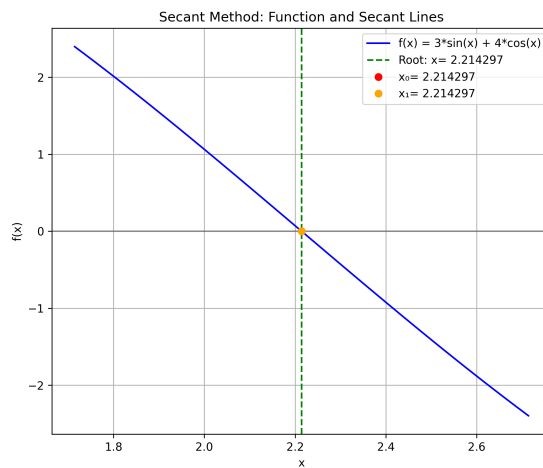
Result: Converged to 2.214297435588180 in 4 iterations **Final Error:** 0.000000000000000 **Convergence Rate:** 5.410312797770888



Secant method from bisection [x9, x10] ($\epsilon = 1e-06$, absolute)

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
1	2.214297435619653	-0.000000000157360	0.000000000031472	N/A	N/A
2	2.214297435588181	0.000000000000001	0.000000000000000	0.000000000031472	N/A

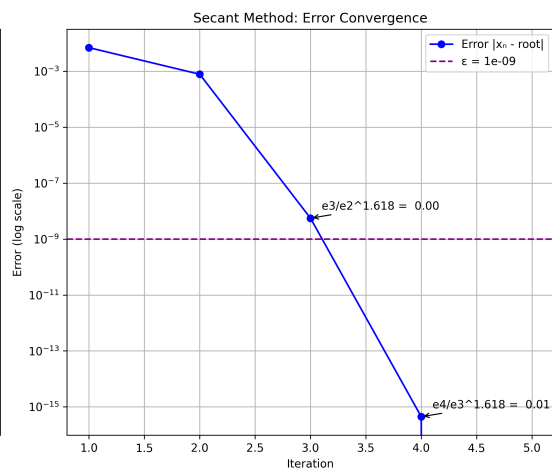
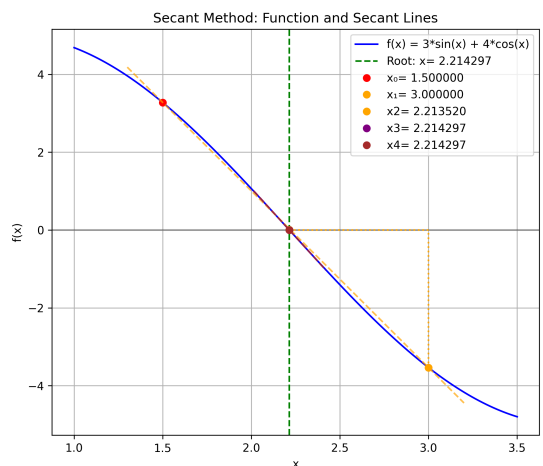
Result: Converged to 2.214297435588181 in 2 iterations Final Error: 0.000000000000000 Convergence Rate: Insufficient data



Secant method from Newton [x2, x3] ($\epsilon = 1e-06$, absolute)

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
1	2.214297435588181	0.000000000000001	0.000000000000000	N/A	N/A

Result: Converged to 2.214297435588181 in 1 iterations Final Error: 0.000000000000000 Convergence Rate: Insufficient data

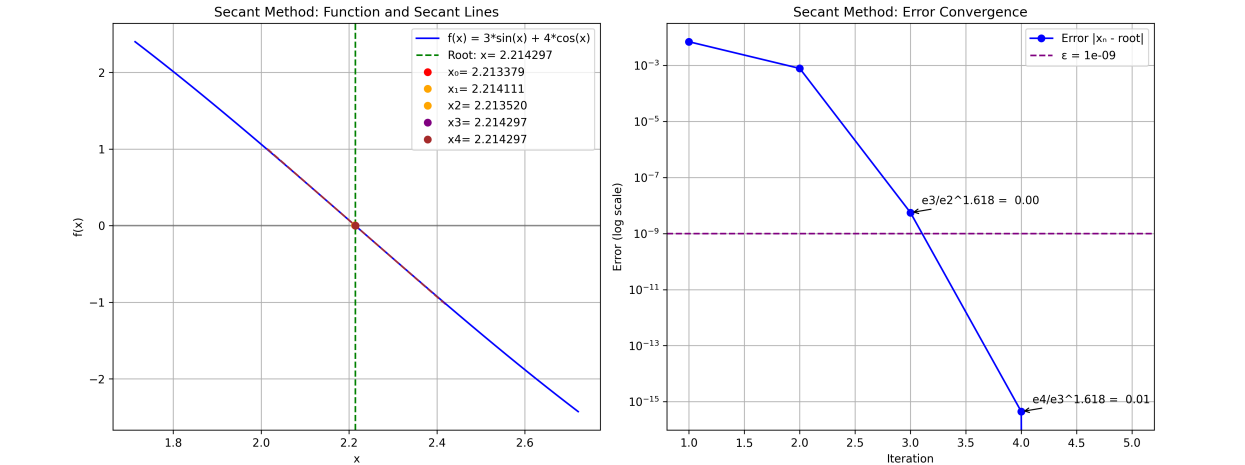


$f(x) = 3\sin(x) + 4\cos(x)$ ($\epsilon = 1e-09$, absolute criterion)

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
1	2.221244731448364	-0.034736199876025	0.006947295860183	N/A	N/A
2	2.213520010199476	0.003887126551967	0.00077425388705	0.007724721248888	N/A
3	2.214297441142124	-0.00000002769715	0.000000005553943	0.00077430942648	2.2142263533919904

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
4	2.214297435588180	0.0000000000000003	0.0000000000000000	0.000000005553944	2.21429743558822
5	2.214297435588181	0.0000000000000001	0.0000000000000000	0.0000000000000000	2.214297435588181

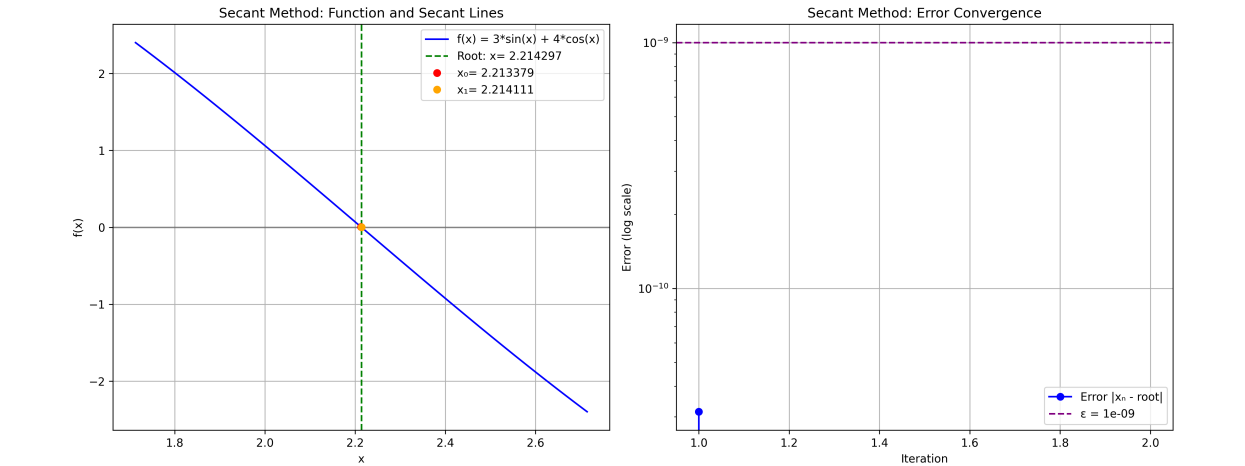
Result: Converged to 2.214297435588181 in 5 iterations **Final Error:** 0.0000000000000000 **Convergence Rate:** 5.410312797770888



Secant method from bisection $[a, b]$ ($\epsilon = 1e-09$, absolute)

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
1	2.221244731448364	-0.034736199876025	0.006947295860183	N/A	N/A
2	2.213520010199476	0.003887126551967	0.00077425388705	0.007724721248888	N/A
3	2.214297441142124	-0.00000002769715	0.000000005553943	0.00077430942648	2.2142263533919904
4	2.214297435588180	0.0000000000000003	0.0000000000000000	0.000000005553944	2.21429743558822
5	2.214297435588181	0.0000000000000001	0.0000000000000000	0.0000000000000000	2.214297435588181

Result: Converged to 2.214297435588181 in 5 iterations **Final Error:** 0.0000000000000000 **Convergence Rate:** 5.410312797770888

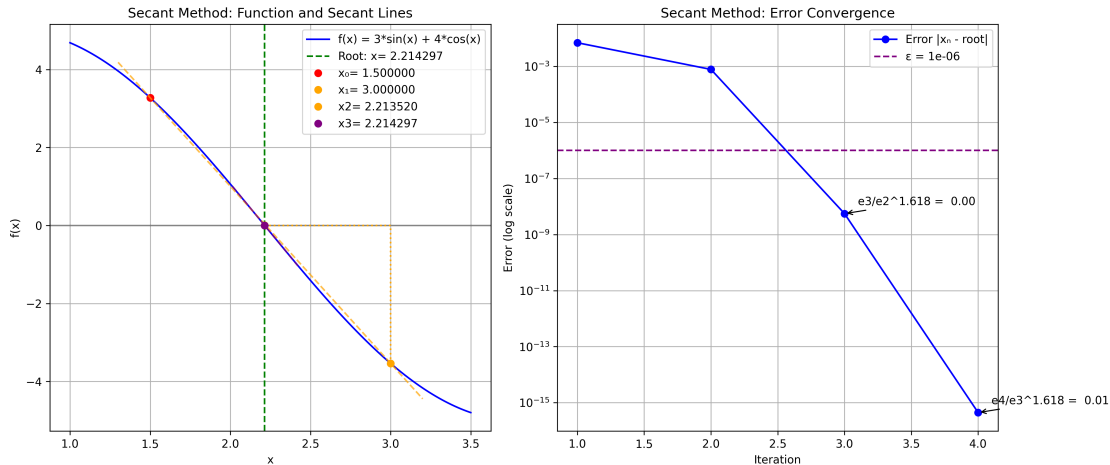


Secant method from bisection $[x_9, x_{10}]$ ($\epsilon = 1e-09$, absolute)

Iteration	x_n	$f(x_n)$	Error	Abs. Error	Aitken
1	2.214297435619653	-0.000000000157360	0.000000000031472	N/A	N/A
2	2.214297435588181	0.0000000000000001	0.0000000000000000	0.000000000031472	N/A

Result: Converged to 2.214297435588181 in 2 iterations **Final Error:** 0.0000000000000000 **Convergence Rate:** Insufficient data

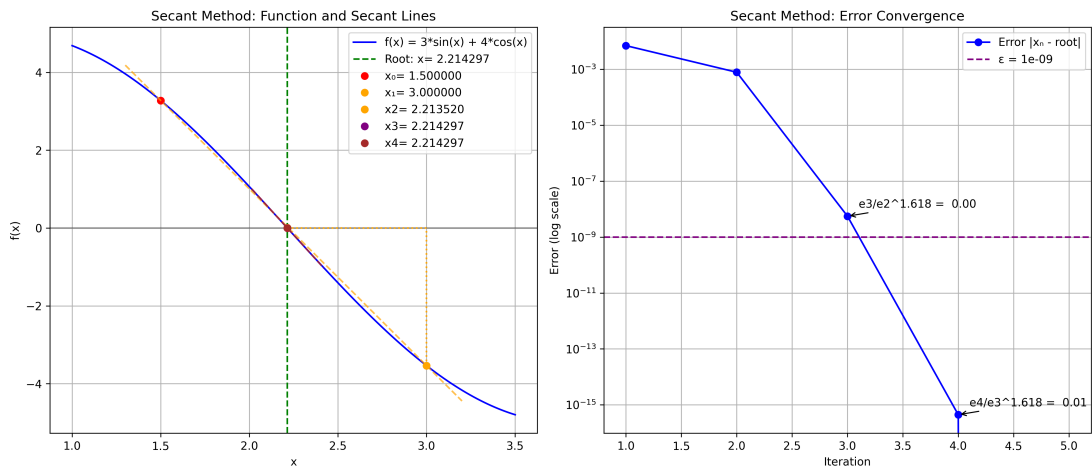
Examples with Relative Error Criterion $|x_{n+1} - x_n|/|x_{n+1}| < \epsilon$



$f(x) = 3\sin(x) + 4\cos(x)$ ($\epsilon = 1e-06$, relative criterion)

Iteration	x_n	$f(x_n)$	Error	Rel. Difference	Aitken
1	2.221244731448364	-0.034736199876025	-4.999879338185890	0.006947295860183	N/A
2	2.213520010199476	0.003887126551967	-4.999998489024489	0.000777425388705	0.003489790565838
3	2.214297441142124	-0.000000027769715	-5.000000000000000	0.000000005553943	0.000351095985662
4	2.214297435588180	0.000000000000003	-5.000000000000000	0.000000000000000	0.000000002508219

Result: Converged to 2.214297435588180 in 4 iterations Final Error: 0.000000000000000 Convergence Rate: 5.410312797770888

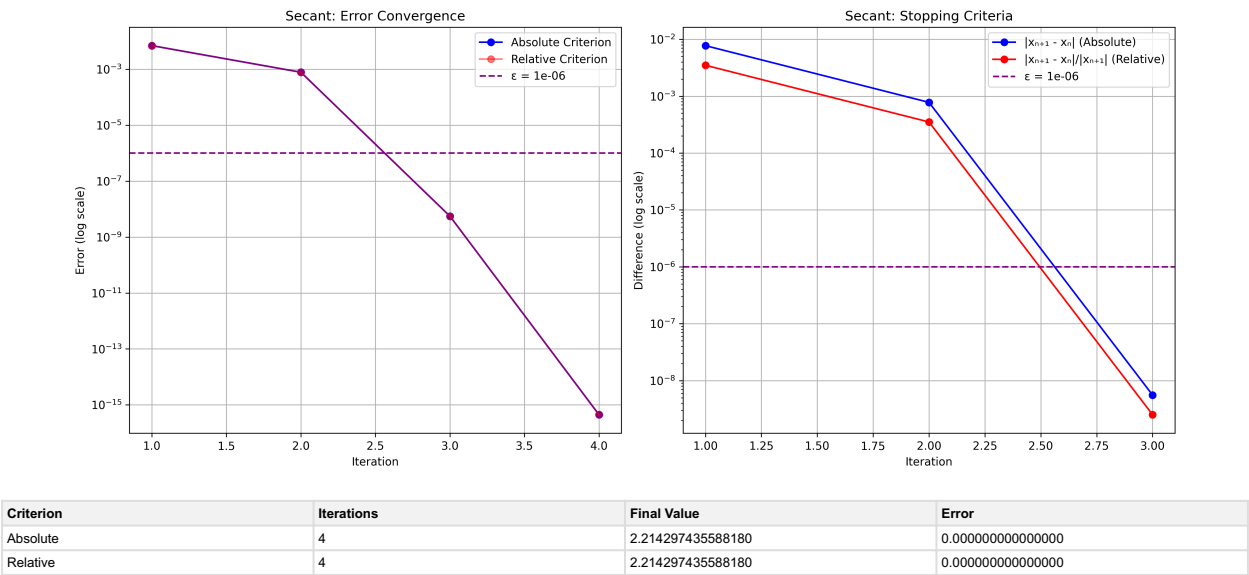


$f(x) = 3\sin(x) + 4\cos(x)$ ($\epsilon = 1e-09$, relative criterion)

Iteration	x_n	$f(x_n)$	Error	Rel. Difference	Aitken
1	2.221244731448364	-0.034736199876025	-4.999879338185890	0.006947295860183	N/A
2	2.213520010199476	0.003887126551967	-4.999998489024489	0.000777425388705	0.003489790565838
3	2.214297441142124	-0.000000027769715	-5.000000000000000	0.000000005553943	0.000351095985662
4	2.214297435588180	0.000000000000003	-5.000000000000000	0.000000000000000	0.000000002508219
5	2.214297435588181	0.000000000000001	-5.000000000000000	0.000000000000000	0.000000000000000

Result: Converged to 2.214297435588181 in 5 iterations Final Error: 0.000000000000000 Convergence Rate: 5.410312797770888

Comparison of Stopping Criteria for Secant Method



4. Fixed-point Method

Mathematical Formula

Transform the equation $f(x) = 0$ into the equivalent form $x = g(x)$, then iterate:

$x_{n+1} = g(x_n)$

Algorithm

- 1. **Input:** Function f , initial guess x_0 , tolerance ϵ
- 2. **Transform:** Define $g(x)$ such that $f(x) = 0 \iff x = g(x)$
- 3. **Initialize:** Set $n = 0$
- 4. **Iterate:**
- 5. Compute $x_{n+1} = g(x_n)$
- 6. If $|error| < \epsilon$, return x_{n+1}
- 7. Set $n = n + 1$
- 8. **Repeat** until convergence or maximum iterations

Convergence Condition

Convergence requires $|g'(r)| < 1$ at the fixed point r : $|e_{n+1}| \leq |g'(\xi)| |e_n|$ where ξ lies between x_n and r .

Implementation

```
def fixed_point_method(f, x0, g, epsilon=1e-6, condition=stopping_condition_2):
    sequence = [x0]
    x = x0

    for k in range(100):
        try:
            x_new = g(x)
            sequence.append(x_new)

            if condition(sequence[k], sequence[k-1], epsilon):
                return sequence, "Second stopping condition (convergence)"

            if abs(x_new) > 1e10 or math.isnan(x_new) or math.isinf(x_new):
                return sequence, "Error: Divergence"

            x = x_new

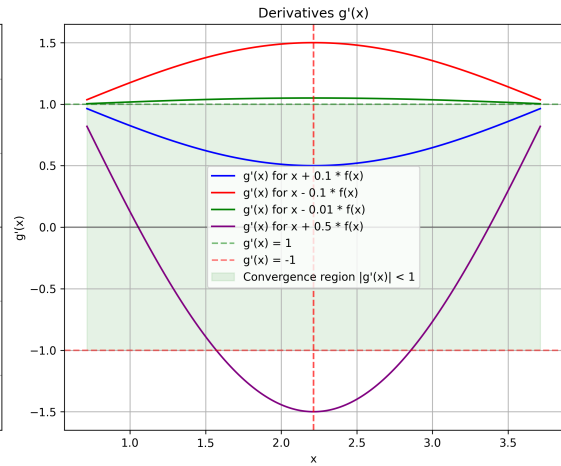
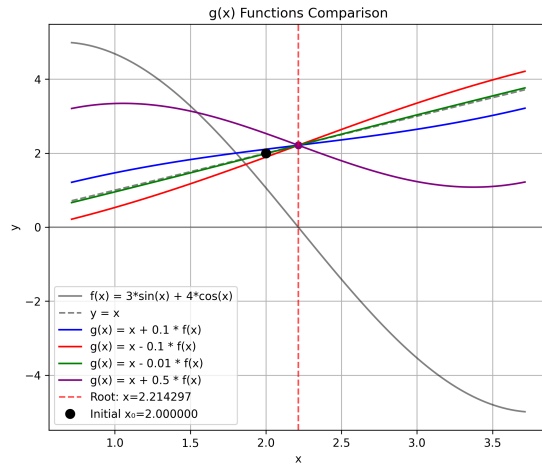
        except Exception:
            return sequence, "Error: Non-convergence"

    return sequence, "First stopping condition (max iterations)"
```

Best $g(x)$ Function

The best $g(x)$ function is: $x + 0.1 * f(x)$

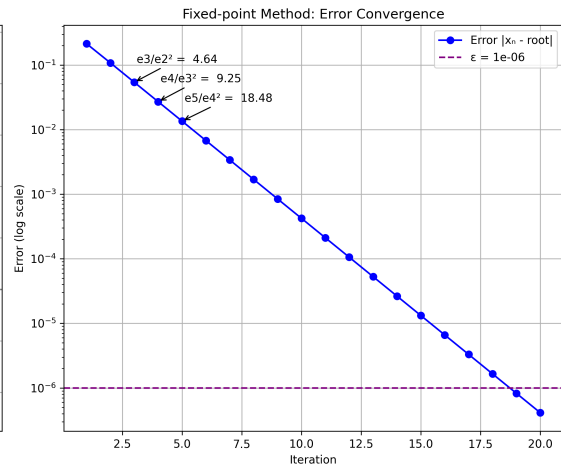
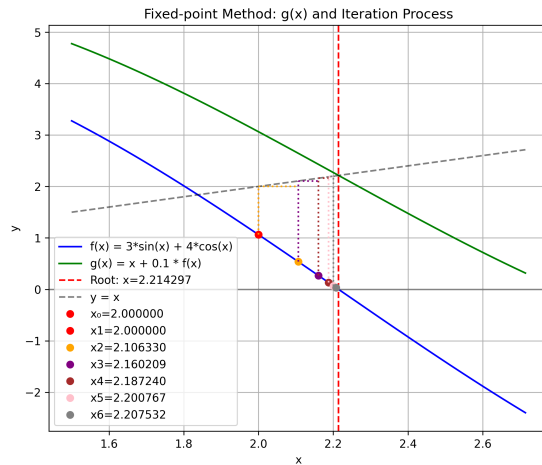
g(x)	Convergence	Error
$x - 0.01 * f(x)$	No	4.868086038083111
$x + 0.01 * f(x)$	No	0.082781966380898
$x - 0.05 * f(x)$	Yes	0.000102981922725
$x + 0.05 * f(x)$	Yes	0.000000610362191
$x - 0.1 * f(x)$	Yes	0.00000000938023
$x + 0.1 * f(x)$	Yes	0.00000000504064
$x - 0.5 * f(x)$	No	4.524410342605131
$x + 0.5 * f(x)$	No	4.524410342605132
$x - 1.0 * f(x)$	No	4.171674420802168
$x + 1.0 * f(x)$	No	0.343884535127269



Explanation:

- The plot shows different $g(x)$ functions and their derivatives
- For convergence, we need $|g'(x)| < 1$ at the root (shaded green area in right plot)
- The best $g(x)$ has its derivative closest to 0 at the root, giving fastest convergence
- Functions with $g'(x)$ outside $[-1, 1]$ at the root will diverge

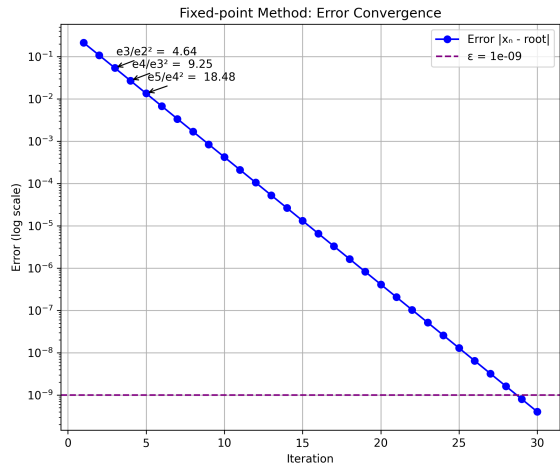
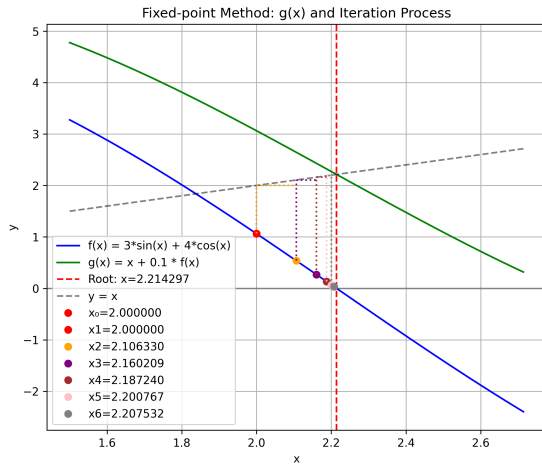
Examples with Absolute Error Criterion $|x_{n+1} - x_n| < \epsilon$



Fixed-point Method ($\epsilon = 0.00001000000000$)

Iteration	x_n	$g(x_n)$	$f(x_n)$	Error	Aitken
1	2.00000000000000	3.063304934288476	1.063304934288475	0.214297435588181	N/A
2	2.106330493428847	2.645117019011117	0.538786525582269	0.107966942159333	N/A
3	2.160209145987074	2.430518748595957	0.270309602608882	0.054088289601106	2.215553422819048
4	2.187240106247963	2.322510246348561	0.135270140100598	0.027057329340218	2.214455585497856
5	2.200767120258022	2.268416632771748	0.067649512513726	0.013530315330159	2.214317241432201
6	2.207532071509395	2.241358633860802	0.033826562351407	0.006765364078786	2.2142999124873044
7	2.210914727744536	2.227828234706653	0.016913506962118	0.003382707843645	2.214297745237162
8	2.212606078440747	2.221062860145878	0.008456781705130	0.001691357147434	2.2142974742954493
9	2.213451756611260	2.217680150991858	0.004228394380597	0.000845678976920	2.214297440426624
10	2.213874596049320	2.215988793680623	0.002114197631303	0.000422839538861	2.2142974361929864
11	2.214086015812450	2.215143114683228	0.001057098870778	0.000211419775730	2.2142974356637812
12	2.214191725699528	2.214720275141808	0.000528549442280	0.000105709888653	2.2142974355976306
13	2.214244580643756	2.214508855365758	0.000264274722002	0.000052854944425	2.2142974355893608
14	2.214271008115956	2.214403145477065	0.000132137361109	0.000026427472225	2.2142974355883296
15	2.214284221852067	2.214350290532635	0.000066068680568	0.000013213736114	2.2142974355881986
16	2.214290828720124	2.214323863060410	0.000033034340286	0.000006606868057	2.2142974355881826
17	2.214294132154152	2.214310649324296	0.000016517170144	0.000003303434029	2.214297435588183
18	2.214295783871167	2.214304042456239	0.000008258585072	0.000001651717014	2.2142974355881826
19	2.214296609729674	2.214300739022209	0.000004129292535	0.000000825858507	2.2142974355881817
20	2.214297022658927	2.214299087305196	0.000002064646268	0.000000412929253	2.214297435588182

Result: Converged **Final Value:** 2.214297022658927 in 20 iterations **Final Error:** 0.000000412929253 **Convergence Rate:** 1.000615115141705

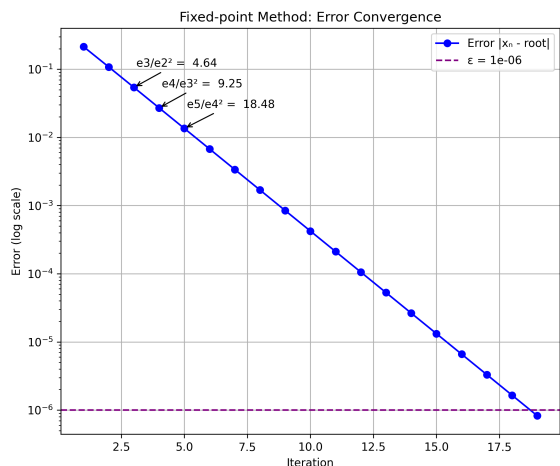
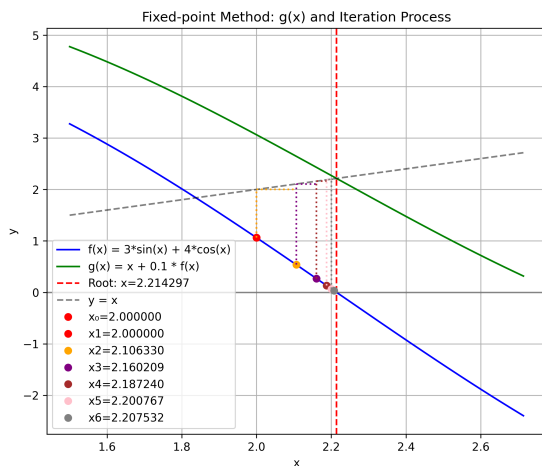


Fixed-point Method ($\epsilon = 0.00000001000000$)

Iteration	x_n	$g(x_n)$	$f(x_n)$	Error	Aitken
1	2.00000000000000	3.063304934288476	1.063304934288475	0.214297435588181	N/A
2	2.106330493428847	2.645117019011117	0.538786525582269	0.107966942159333	N/A
3	2.160209145987074	2.430518748595957	0.270309602608882	0.054088289601106	2.215553422819048
4	2.187240106247963	2.322510246348561	0.135270140100598	0.027057329340218	2.214455585497856
5	2.200767120258022	2.268416632771748	0.067649512513726	0.013530315330159	2.214317241432201
6	2.207532071509395	2.241358633860802	0.033826562351407	0.006765364078786	2.2142999124873044
7	2.210914727744536	2.227828234706653	0.016913506962118	0.003382707843645	2.214297745237162
8	2.212606078440747	2.221062860145878	0.008456781705130	0.001691357147434	2.2142974742954493
9	2.213451756611260	2.217680150991858	0.004228394380597	0.000845678976920	2.214297440426624
10	2.213874596049320	2.215988793680623	0.002114197631303	0.000422839538861	2.2142974361929864
11	2.214086015812450	2.215143114683228	0.001057098870778	0.000211419775730	2.2142974356637812
12	2.214191725699528	2.214720275141808	0.000528549442280	0.000105709888653	2.2142974355976306
13	2.214244580643756	2.214508855365758	0.000264274722002	0.000052854944425	2.2142974355893608
14	2.214271008115956	2.214403145477065	0.000132137361109	0.000026427472225	2.2142974355883296
15	2.214284221852067	2.214350290532635	0.000066068680568	0.000013213736114	2.2142974355881986
16	2.214290828720124	2.214323863060410	0.000033034340286	0.000006606868057	2.2142974355881826
17	2.214294132154152	2.214310649324296	0.000016517170144	0.000003303434029	2.214297435588183
18	2.214295783871167	2.214304042456239	0.000008258585072	0.000001651717014	2.2142974355881826
19	2.214296609729674	2.214300739022209	0.000004129292535	0.000000825858507	2.2142974355881817
20	2.214297022658927	2.214299087305196	0.000002064646268	0.000000412929253	2.214297435588182
21	2.214297229123554	2.214298261446689	0.000001032323135	0.000000206464627	2.214297435588181
22	2.214297332355867	2.214297848517435	0.000000516161568	0.000000103232313	2.214297435588181
23	2.214297383972024	2.214297642052808	0.000000258080784	0.000000051616157	2.214297435588181
24	2.214297409780102	2.214297538820495	0.000000129040393	0.000000025808078	2.214297435588181
25	2.214297422684142	2.214297487204338	0.000000064520197	0.000000012904039	2.2142974355881826
26	2.214297429136161	2.214297461396259	0.000000032260098	0.000000006452019	2.2142974355881817
27	2.214297432362171	2.214297448492220	0.000000016130049	0.000000003226010	2.214297435588182
28	2.214297433975176	2.214297442040201	0.000000008065025	0.000000001613005	2.214297435588181
29	2.214297434781678	2.214297438814191	0.000000004032513	0.000000000806502	2.214297435588181
30	2.214297435184930	2.214297437201187	0.000000002016257	0.000000000403251	2.2142974355881826

Result: Converged **Final Value:** 2.214297435184930 in 30 iterations **Final Error:** 0.00000000403251 **Convergence Rate:** 1.000395431162525

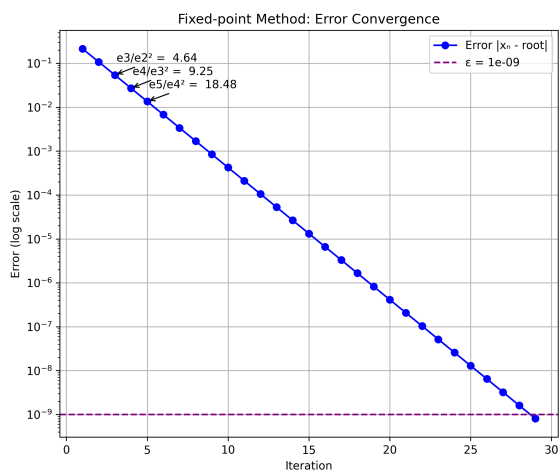
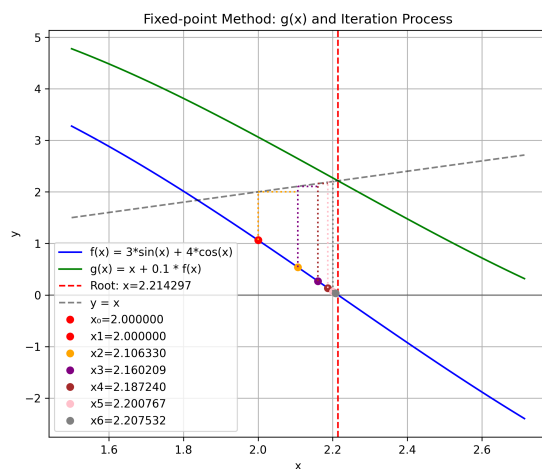
Examples with Relative Error Criterion $|x_{n+1} - x_n| / |x_{n+1}| < \epsilon$



Fixed-point Method with Relative Criterion ($\epsilon = 0.000001000000000$)

Iteration	x_n	$g(x_n)$	$f(x_n)$	Error	Rel. Difference	Aitken
1	2.000000000000000	3.063304934288476	1.063304934288475	0.214297435588181	N/A	N/A
2	2.106330493428847	2.645117019011117	0.538786525582269	0.107966942159333	0.050481391101999	N/A
3	2.160209145987074	2.430518748595957	0.270309602608882	0.054088289601106	0.024941405631170	2.215553422819048
4	2.187240106247963	2.322510246348561	0.135270140100598	0.027057329340218	0.012358478698188	2.214455585497856
5	2.200767120258022	2.268416632771748	0.067649512513726	0.013530315330159	0.006146499502625	2.214317241432201
6	2.207532071509395	2.241358633860802	0.033826562351407	0.006765364078786	0.003064486056027	2.2142999124873044
7	2.210914727744536	2.227828234706653	0.016913506962118	0.003382707843645	0.001529980416111	2.214297745237162
8	2.212606078440747	2.221062860145878	0.008456781705130	0.001691357147434	0.000764415642121	2.2142974742954493
9	2.213451756611260	2.217680150991858	0.004228394380597	0.000845678976920	0.000382063068683	2.214297440426624
10	2.213874596049320	2.215988793680623	0.002114197631303	0.000422839538861	0.000190995207594	2.2142974361929864
11	2.214086015812450	2.215143114683228	0.001057098870778	0.000211419775730	0.000095488504792	2.2142974356637812
12	2.214191725699528	2.214720275141808	0.000528549442280	0.000105709888653	0.000047741975481	2.2142974355976306
13	2.214244580643756	2.214508855365758	0.000264274722002	0.000052854944425	0.000023870418241	2.2142974355893608
14	2.214271008115956	2.214403145477065	0.000132137361109	0.000026427472225	0.000011935066712	2.2142974355883296
15	2.214284221852067	2.214350290532635	0.000066068680568	0.000013213736114	0.000005967497750	2.2142974355881986
16	2.214290828720124	2.214323863060410	0.000033034340286	0.000006606868057	0.000002983739973	2.2142974355881826
17	2.214294132154152	2.214310649324296	0.000016517170144	0.000003303434029	0.000001491867761	2.214297435588183
18	2.214295783871167	2.214304042456239	0.000008258585072	0.000001651717014	0.000000745933324	2.2142974355881826
19	2.214296609729674	2.214300739022209	0.000004129292535	0.000000825858507	0.000000372966523	2.2142974355881817

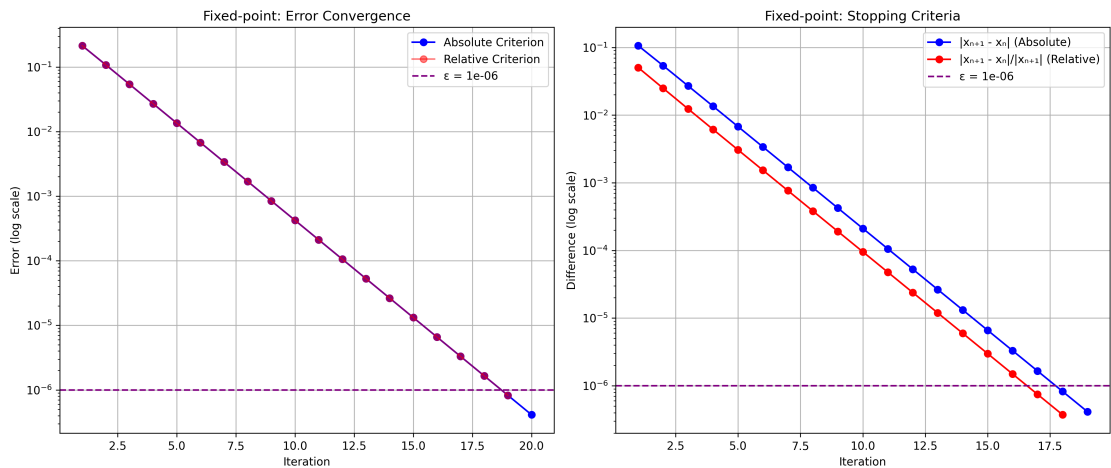
Result: Converged **Final Value:** 2.214296609729674 in 19 iterations **Final Error:** 0.000000825858507 **Convergence Rate:** 1.000651298408151

Fixed-point Method with Relative Criterion ($\epsilon = 0.000000010000000$)

Iteration	x_n	$g(x_n)$	$f(x_n)$	Error	Rel. Difference	Aitken
1	2.000000000000000	3.063304934288476	1.063304934288475	0.214297435588181	N/A	N/A
2	2.106330493428847	2.645117019011117	0.538786525582269	0.107966942159333	0.050481391101999	N/A
3	2.160209145987074	2.430518748595957	0.270309602608882	0.054088289601106	0.024941405631170	2.215553422819048
4	2.187240106247963	2.322510246348561	0.135270140100598	0.027057329340218	0.012358478698188	2.214455585497856
5	2.200767120258022	2.268416632771748	0.067649512513726	0.013530315330159	0.006146499502625	2.214317241432201
6	2.207532071509395	2.241358633860802	0.033826562351407	0.006765364078786	0.003064486056027	2.2142999124873044
7	2.210914727744536	2.227828234706653	0.016913506962118	0.003382707843645	0.001529980416111	2.214297745237162
8	2.212606078440747	2.221062860145878	0.008456781705130	0.001691357147434	0.000764415642121	2.2142974742954493
9	2.213451756611260	2.217680150991858	0.004228394380597	0.000845678976920	0.000382063068683	2.214297440426624
10	2.213874596049320	2.215988793680623	0.002114197631303	0.000422839538861	0.000190995207594	2.2142974361929864
11	2.214086015812450	2.215143114683228	0.001057098870778	0.000211419775730	0.000095488504792	2.2142974356637812
12	2.214191725699528	2.214720275141808	0.000528549442280	0.000105709888653	0.000047741975481	2.2142974355976306
13	2.214244580643756	2.214508855365758	0.000264274722002	0.000052854944425	0.000023870418241	2.2142974355893608
14	2.214271008115956	2.214403145477065	0.000132137361109	0.000026427472225	0.000011935066712	2.2142974355883296
15	2.214284221852067	2.214350290532635	0.000066068680568	0.000013213736114	0.000005967497750	2.2142974355881986
16	2.214290828720124	2.214323863060410	0.000033034340286	0.000006606868057	0.000002983739973	2.2142974355881826
17	2.214294132154152	2.214310649324296	0.000016517170144	0.000003303434029	0.000001491867761	2.214297435588183
18	2.214295783871167	2.214304042456239	0.000008258585072	0.000001651717014	0.000000745933324	2.2142974355881826
19	2.214296609729674	2.214300739022209	0.000004129292535	0.000000825858507	0.000000372966523	2.2142974355881817
20	2.214297022658927	2.214299087305196	0.000002064646268	0.000000412929253	0.000000186483227	2.214297435588182
21	2.214297229123554	2.214298261446689	0.000001032323135	0.000000206464627	0.000000093241605	2.214297435588181
22	2.21429733235867	2.214297848517435	0.000000516161568	0.000000103232313	0.000000046620800	2.214297435588181
23	2.214297383972024	2.214297642052808	0.000000258080784	0.000000051616157	0.000000023310400	2.214297435588181
24	2.214297409780102	2.214297538820495	0.000000129040393	0.000000025808078	0.000000011655200	2.214297435588181
25	2.214297422684142	2.214297487204338	0.000000064520197	0.000000012904039	0.000000005827600	2.2142974355881826
26	2.214297429136161	2.214297461396259	0.000000032260098	0.000000006452019	0.000000002913800	2.2142974355881817
27	2.214297432362171	2.214297448492220	0.000000016130049	0.000000003226010	0.000000001456900	2.214297435588182
28	2.214297433975176	2.214297442040201	0.000000008065025	0.000000001613005	0.000000000728450	2.214297435588181
29	2.214297434781678	2.214297438814191	0.000000004032513	0.000000000806502	0.000000000364225	2.214297435588181

Result: Converged **Final Value:** 2.214297434781678 in 29 iterations **Final Error:** 0.00000000806502 **Convergence Rate:** 1.000410076761137

Comparison of Stopping Criteria for Fixed-point Method



Criterion	Iterations	Final Value	Error
Absolute	20	2.214297022658927	0.000000412929253
Relative	19	2.214296609729674	0.000000825858507

5. Aitken's Acceleration

Mathematical Formula

For a sequence $\{p_n\}$ converging to limit p , Aitken's Δ^2 method produces:

$$\hat{p}_n = p_n - (\Delta p_n)^2 / \Delta^2 p_n$$

where: $\Delta p_n = p_{n+1} - p_n$ (forward difference) - $\Delta^2 p_n = \Delta p_{n+1} - \Delta p_n = p_{n+2} - 2p_{n+1} + p_n$ (second difference)

Expanded form: $\hat{p}_n = p_n - \frac{\Delta p_n^2}{p_{n+2} - 2p_{n+1} + p_n}$

Algorithm

1. **Input:** Sequence $\{p_0, p_1, p_2, \dots\}$, tolerance ϵ
2. **For each** $n \geq 0$ where p_{n+2} exists:
3. Compute $\Delta^2 p_n = p_{n+2} - 2p_{n+1} + p_n$
4. If $|\Delta^2 p_n| < \text{small threshold}$, skip (avoid division by zero)
5. Compute $\hat{p}_n = p_n - \frac{\Delta p_n^2}{\Delta^2 p_n}$
6. **Return** accelerated sequence

Implementation

```
def aitken_acceleration(sequence, epsilon=1e-6):
    if len(sequence) < 3:
        return sequence, "Short sequence - minimum 3 elements needed"

    aitken_sequence = []

    for i in range(len(sequence) - 2):
        p0, p1, p2 = sequence[i], sequence[i + 1], sequence[i + 2]

        denominator = p0 + p2 - 2 * p1
        if abs(denominator) < 1e-19:
            accelerated = p2
        else:
            accelerated = p0 - ((p1 - p0)**2 / denominator)

        aitken_sequence.append(accelerated)

    return aitken_sequence, "First stopping condition (max iterations)"
```

6. Real-time Aitken's Acceleration with Fixed-point

Mathematical Formula

This method applies Aitken's acceleration dynamically during the Fixed-point process:

1. Generate a few iterations using Fixed-point: $x_{n+1} = g(x_n)$
2. When three consecutive points are available, compute the Aitken accelerated value: $\hat{x}_n = x_n - \frac{\Delta x_n^2}{\Delta^2 x_n}$
3. Replace the current approximation with the accelerated value and continue

Algorithm

1. **Input:** Function f , initial guess x_0 , tolerance ϵ , max iterations
2. **Initialize:** Set $n = 0$, sequence = $[x_0]$
3. **Iterate:**
4. Compute next Fixed-point value: $x_{n+1} = g(x_n)$
5. Add x_{n+1} to sequence
6. If at least 3 points exist in sequence, apply Aitken acceleration
7. If accelerated value meets convergence criterion, return it
8. Otherwise, continue with the next Fixed-point
9. **Return** final value and performance metrics

Implementation

```
def fixed_point_with_aitken(f, x0, g, epsilon=1e-6, max_iterations=20, condition=stopping_condition_2):
    regular_sequence = [x0]
    aitken_values = {}

    x = x0

    method_info = {
        'converged': False,
        'convergence_type': None,
        'final_value': None,
        'iterations': 0,
        'aitken_applications': 0
    }

    for k in range(max_iterations):
        x_new = g(x)
        regular_sequence.append(x_new)
```

```
method_info['iterations'] = k + 1

if k >= 2:
    p0, p1, p2 = regular_sequence[k - 2], regular_sequence[k - 1], regular_sequence[k]
    denominator = p2 - 2 * p1 + p0
    if abs(denominator) > 1e-19:
        aitken_val = p0 - (p1 - p0)**2 / denominator
        aitken_values[k] = aitken_val
        method_info['aitken_applications'] += 1
        if (k - 1) in aitken_values:
            if condition(aitken_val, aitken_values[k - 1], epsilon):
                method_info['converged'] = True
                method_info['convergence_type'] = 'Aitken acceleration'
                method_info['final_value'] = aitken_val
                break
    x = x_new

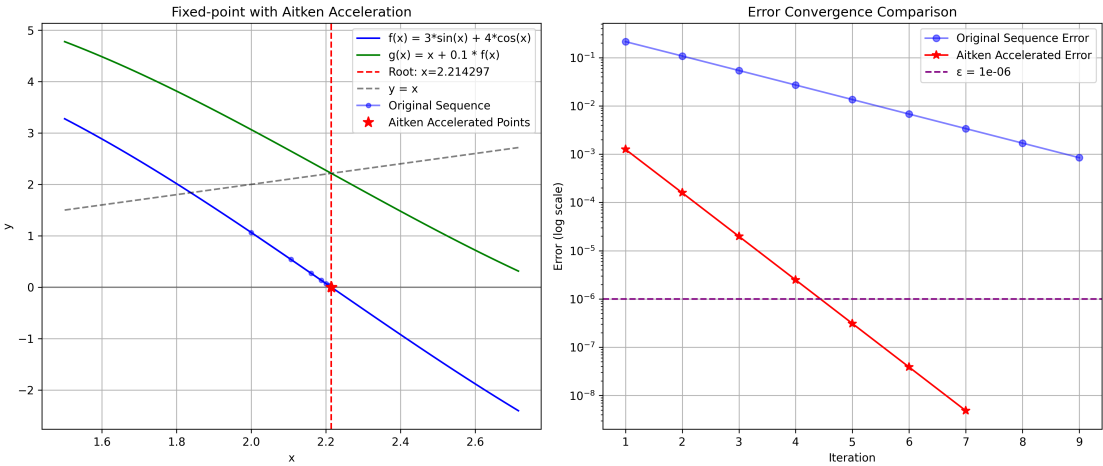
if condition(regular_sequence[-1], regular_sequence[-2], epsilon):
    method_info['converged'] = True
    method_info['convergence_type'] = 'Regular iteration'
    method_info['final_value'] = regular_sequence[-1]
    break

if abs(x) > 1e10:
    method_info['convergence_type'] = 'Diverged'
    method_info['final_value'] = x
    break

if not method_info['converged'] and method_info['convergence_type'] != 'Diverged':
    method_info['convergence_type'] = 'Max iterations reached'
    if aitken_values:
        last_aitken_key = max(aitken_values.keys())
        method_info['final_value'] = aitken_values[last_aitken_key]
    else:
        method_info['final_value'] = regular_sequence[-1]

return regular_sequence, aitken_values, method_info
```

Examples with Real-time Aitken Acceleration



Fixed-point with Real-time Aitken ($\epsilon = 0.000001000000000$)

Iteration	x_n	$f(x_n)$	Error	Aitken Applied	Acceleration Effect
1	2.000000000000000	1.063304934288475	0.214297435588181	Yes	N/A
2	2.106330493428847	0.538786525582269	0.107966942159333	Yes	682.687346342903993x faster
3	2.160209145987074	0.270309602608882	0.054088289601106	Yes	Excellent
4	2.187240106247963	0.135270140100598	0.027057329340218	Yes	Excellent
5	2.200767120258022	0.067649512513726	0.013530315330159	Yes	Excellent
6	2.207532071509395	0.033826562351407	0.006765364078786	Yes	Excellent
7	2.210914727744536	0.016913506962118	0.003382707843645	Yes	Excellent
8	2.212606078440747	0.008456781705130	0.001691357147434	No	N/A
9	2.213451756611260	0.004228394380597	0.000845678976920	No	N/A

Result: Converged to 2.214297474295448 in 8 iterations **Final Error:** 0.000000038707267 **Aitken Applications:** 6 **Acceleration Effect:** Significant improvement over basic Fixed-point

Performance Analysis

Summary Table

Method	Avg Iterations	Avg Time (ms)	Avg Error	Convergence Rate	Success Rate
Bisection	25.500000000000000	0.034868717193604	0.000000466130685	1.321536756744824	40.000000000000000%
Newton	5.500000000000000	0.007688999176025	0.000000000000000	3.004455781252174	40.000000000000000%
Secant	3.555555555555555	0.003602769639757	0.000000000000000	5.410312797770888	90.000000000000000%
Fixed-point	24.500000000000000	0.022768974304199	0.000000309999379	1.000517980368379	40.000000000000000%
Bisection + Aitken	23.500000000000000	0.016093254089355	0.000000406817079	0.228259968535387	40.000000000000000%
Fixed-point + Aitken	22.500000000000000	0.011503696441650	0.000000000000001	1.009092223122644	40.000000000000000%
Real-time Aitken	9.750000000000000	0.026226043701172	0.000449276454738	1.001410231401462	40.000000000000000%
Periodic Aitken	8.000000000000000	0.017523765563965	0.000000000000015	1.931300996100426	20.000000000000000%